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United States Patent	12411397
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Neibert; Dana

Cable protector and securement system

Abstract

A cable protector and securement system comprising a cable protector device configured attach to a camera for preventing a connector of a cable from becoming unintentionally disconnected from camera and/or damaged when the cable is installed to camera. In certain embodiments, the cable protector device may include a cable securing cartridge configured for securement of the cable in a manner which restrains movement of the cable, and a cable connector protector coupled to the cable securing cartridge.

Inventors: **Neibert; Dana (Coronado, CA)**

Applicant: **Neibert; Dana** (Coronado, CA)

Family ID: **96950441**

Appl. No.: **18/511812**

Filed: **November 16, 2023**

Publication Classification

Int. Cl.: **G03B17/56** (20210101); **H01R13/58** (20060101)

U.S. Cl.:

CPC **G03B17/561** (20130101); **H01R13/5804** (20130101);

Field of Classification Search

CPC: G03B (17/56-566)

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Primary Examiner: Rhodes, Jr.; Leon W

Attorney, Agent or Firm: Plager Schack LLP

Background/Summary

BACKGROUND
(1) The present disclosure relates generally to camera cable protection systems.
(2) Photographers may typically use tether cables, (typically long USB cables) to connect their camera to a computer, for example, for seeing live images from their camera on the computer screen in real time. However, such tether cables are expensive and subject to easily becoming damaged. Moreover, the tether cable can easily disconnect from the camera during normal use. The As such, there is a need for an improved system that addresses these issues.
SUMMARY
(3) According to various embodiments, disclosed is a cable protector and securement system comprising a cable protector device configured attach to a camera for preventing a connector of the cable from becoming unintentionally disconnected from camera and/or damaged when the cable is installed to camera. In certain embodiments, the cable protector device may include a cable securing cartridge configured for securement of the cable in a manner which restrains movement of the cable, and a cable connector protector coupled to the cable securing cartridge. In certain embodiments, the cable connector protector includes at least one wall configured to shield the

connector of the cable when the connector is plugged into a connector receiving port of the camera, and when the cable protector device is attached to the camera.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The detailed description of some embodiments of the invention will be made below with reference to the accompanying figures, wherein the figures disclose one or more embodiments of the present invention.

(2) FIG. **1** is a side perspective view of a cable protector device of a cable securement system, the device comprising a cable securing cartage and a cable connector protector, according to various embodiments.

(3) FIG. **2** is a front perspective view showing the protector device of FIG. **1** attached to a camera, and in use for protection of a cable installed to the camera according to various embodiments.

(4) FIG. **3** is an enlarged detailed bottom perspective view of the protector device and camera, shown in FIG. **2**.

(5) FIG. **4** is a bottom exploded view of the protector device with cable secured thereto, according to various embodiments.

(6) FIG. **5** is a detail cross-sectional view on line **5-5** of FIG. **3**.

(7) FIG. **6** is an exploded perspective view of the cable securing cartridge of the cable protector device, according to various embodiments.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(8) According to various embodiments as depicted in FIGS. **1-6** disclosed is a cable protector and securement system comprising a cable protector device **10** configured attach to a camera **11** for preventing a connector **24**B of a cable **24**A from becoming unintentionally disconnected from camera **11** and/or damaged when cable **24**A is installed to camera **11**. In certain embodiments, cable **24**A may be a USB and/or data cable with connector **24**B being a USB plug and/or data cable plug, but is not limited to these options. Cable **24**A is installed to camera **11** by plugging connector **24**B into a connector receiving port **11**A of the camera.

(9) In certain embodiments, device **10** comprises a cable securing cartridge **22** configured to secure cable **24**A to camera **11**. In further embodiments, device **10** comprises a cable connector protector **20** which includes at least one wall configured to shield cable connector **24**B when plugged into the connector receiving port **11**A. In certain embodiments, cable connector protector **20** and cable securing cartridge **22** are coupled to one another. As such, device **10** prevents connector **24**B from detachment and/or damage by preventing cable **24**A from being pulled, while shielding connector **24**B from impact.

(10) In certain embodiments, an extension arm **21** is provided between cable securing cartridge **22** and cable connector protector **20**. Extension arm **21** is configured to position cable connector protector **20** proximate connector receiving port **11**A of the camera, such that the cable connector protector **20** creates at least one shielding wall alongside connector **24**B when connector **24**B is plugged in. In certain embodiments, device **10** is configured to mount to an underside wall **16** of camera **11**, wherein connector receiving port **11**A is provided on a side of the camera extending substantially perpendicular to the underside wall of the camera. As such, arm **21** positions cable connector protector **20** in a substantial perpendicular alignment to cable securing cartridge **22**. This allows cable **24**A to extend around adjacent walls of camera **11** while held in place via cable securing cartridge **22** for optimal stability.

(11) In some embodiments, device **10** is configured to mount to underside wall **16** of camera **11** via a camera mounting plate **12**. In some embodiments, cartridge **22** may be releasably attachable to the mounting plate, as will be described. In certain embodiments, extension arm **21**/cable connector

protector **20** is an attachment component of the mounting plate **12**. In one embodiment, mounting plate **12** may be attached to underside wall **16** via fastener(s) **18** (i.e., screws) engaged through at least one mounting plate slot **14** and a corresponding internally threaded screw hole **16**A in camera underside wall **16**. In further embodiments, extension arm **21** of cable protector **20** may be assembled to mounting plate **12** via fasteners **18** engaged through corresponding slots and screw holes in arm **21** and plate **12**, respectively (see FIG. **4**). In other embodiments, mounting plate **12** and extension arm **21**/cable protector **20** may be an integral unit.

(12) In certain embodiments, connector protector **20** comprises an elongated plate which is sized and structured to shield connector **24**B along a first major side face of connector **24**B, as shown. In some embodiments, connector protector **20** may include one or more openings **20**A, for heat dissipation, but not necessarily so. In some embodiments, camera **11** may include a port tab **11**B configured to extend along a second major side face opposite the first side face of connector **24**B when connector **24**B is plugged in, wherein connector protector **20** is configured to position opposite port tab **11**B with respect to connector receiving port **11**A. As such, connector protector **20** forms a protective trench for connector **24**B in conjunction with port tab **11**B. It shall be appreciated that while cable connector protector **20** is illustrated as including a single elongated plate, other shielding structures may be provided in alternate embodiments depending on factors such as the camera design, user and/or manufacturer needs, user and/or manufacturer preferences, etc. For example, in a camera not including port tab **11**B, connector protector **20** may include multiple shielding walls for protectively surrounding connector **24**B.

(13) In embodiments, cable securing cartridge **22** is configured to hold cable **24**A in place and prevent the cable from sliding or shifting within cartridge **22**. In further embodiments cartridge **22** is configured to releasably attach to mounting plate **12**. In some embodiments as best depicted in FIG. **6**, cartridge **22** may comprise a main body including bottom wall **22**D and side walls **22**E extending upwards from bottom wall **22**D to define a hollow passage space **22**C through which cable **24**B may pass. In some embodiments, the main body of cartridge **22** may be generally planar. In some further embodiments, the main body of cartridge **22** may also be elongated as depicted in the figures, but not necessarily so. In further embodiments, front and rear knobs **25**A, **25**B may form partial front and rear walls of the cartridge with constricted openings **23** leading into space **22**C. In some embodiments, cartridge **22** may further comprise a cover **22**A configured to releasably attach over side walls **22**E and knobs **25**A, **25**B, opposite bottom wall **22**D to enclose space **22**C. In one embodiment, cover **22**A may include side tabs **22**F configured to slidable engage with recessed tracks **22**G provided on an inner side of side walls **22**E. In further embodiments, a bottom side of cover **22**A may include a bump **27** configured to engage within a receiving dimple **25**A on a top side of rear knob **25**B for guiding the lateral position of the cover. As such, cover **22**A may be easily attachable and detachable component of cartridge **22**.

(14) In certain embodiments, constricted openings **23** provide entrance and exit points for the cable **24**B into and out of passage space **22**C, as shown. Additionally, at least one internal cable engagement knob **22**B is provided within passage space **22**C, and defines an internal cable constricting zone **23**A within the cartridge. In further embodiments, constricted openings **23** and cable constricting zone **23**A provide a passage space slightly larger than the cable's diameter, to allow the cable to pass yet restrict its radial (i.e., traverse) movement. Additionally, cable constricting zone **23**A is transversely offset from constricted openings **23** forcing cable **24**A to take a crooked path within cartridge **22**. This further restricts the cables movement within the cartridge by increasing frictional forces against lateral movement. As such, cable **24**A is restricted from shifting into and out of cartridge **22**, and is effectively secured within cartridge when cover **22**A is installed.

(15) In some embodiment, cartridge **22** is configured to securely and releasably attach to mounting plate **12** when cartridge **22** is enclosed via cover **22**A. In one embodiment, cartridge **22** may comprise a latched peg **27** configured to lockably engage to a latch receiving cavity **27**A formed at

a top side of mounting plate **12**. In some embodiments, latched peg **27**A may extend from a top side opening within front knob **25**A, and may further include a release lever **26** extending from a front side opening of front knob **25**A, as shown. In some embodiments, cover **22**A includes a cutout section which leaves latched peg **27**A exposed at the top surface of cartridge **22**. Release lever **26** is configured to cause latched peg **27** to pull frontwards and disengage from latch receiving cavity, upon depression of release lever **26**. As such, cartridge **22** may be coupled to camera **11**/mounting plate **12**, and released therefrom via latched peg **27**A and release lever **26**.
(16) As such, a user may initially install device **10** to camera **11** by screwing mounting plate **12** to the underside wall of camera **11**. This positions cable connector protector **20** alongside cable port **11**A. In some embodiments, cable connector protector **20** may be attached to mounting plate **12** via fasteners/screws prior to installation of the plate to the camera. In other embodiments, cable connector protector **20** may be pre-attached to mounting plate **12**. Thereafter, device **10** may be used to secure cable **24**A anytime the cable is installed to the camera, without additional hardware or tools needed. In some embodiments, device **10** may be pre-attached to camera **11**.
(17) With mounting plate **12** installed to camera **11**, the user may insert cable **24**A into cartridge **22** and plug cable connector **24**B into cable port **11**A of the camera. In some embodiments, cartridge **22** may first be removed from mounting plate **12** and cover **22**A may be removed from cartridge **22**. Cable **24**A may then be inserted into cartridge **22** and snaked around knobs **25**A, **22**B, and **25**B. Cover **22**A may then be closed over cartridge **22**, and the cartridge may be attached to the mounting plate with cable connector **24**B plugged into connector port **11**A before or after attachment of the cartridge. The resulting assembly keeps the cable securely weaved into the cartridge and the cable connecter protected.
(18) Thus, the disclosed subject matter provides a tool-free system that prevents the cable from unintentionally disconnecting from the camera and also protects the fragile USB connector from damage.
(19) It shall be appreciated that the disclosed device and system can have multiple configurations in different embodiments, and may be adapted for different camera designs in alternate embodiments. In certain embodiments, components of device **10** may be fabricated out of aluminum and/or plastic. It shall be appreciated, however, that device **10** may comprise any alternative known materials in the field and be of any color, size, and/or dimensions. It shall be appreciated that device **10** may be manufactured and assembled using any known techniques in the field.
(20) It shall be understood that the orientation or positional relationship indicated by terms such as “upper”, “lower”, “front”, “rear”, “left”, “right”, “top”, “bottom”, “inside”, “outside” is based on the orientation or positional relationship shown in the accompanying drawings, which is only for convenience and simplification of describing the disclosed subject matter, rather than indicating or implying that the indicated device or element must have a specific orientation or are constructed and operated in a specific orientation, and therefore should not be construed as a limitation of the present invention.
(21) As used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Where only one item is intended, the term “one” or similar language is used. Also, as used herein, the terms “has”, “have”, “having”, “with” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.
(22) The constituent elements of the disclosed device and system listed herein are intended to be exemplary only, and it is not intended that this list be used to limit the device of the present application to just these elements. Persons having ordinary skill in the art relevant to the present disclosure may understand there to be equivalent elements that may be substituted within the present disclosure without changing the essential function or operation of the device. Terms such as ‘approximate,’ ‘approximately,’ ‘about,’ etc., as used herein indicate a deviation of within +/−10%. Relationships between the various elements of the disclosed device as described herein are

presented as illustrative examples only, and not intended to limit the scope or nature of the relationships between the various elements. Persons of ordinary skill in the art may appreciate that numerous design configurations may be possible to enjoy the functional benefits of the inventive systems. Thus, given the wide variety of configurations and arrangements of embodiments of the present invention the scope of the invention is reflected by the breadth of the claims below rather than narrowed by the embodiments described above.

Claims

1. A cable protector device for protecting a cable attached to a camera, the cable protector device comprising: a cable securing cartridge configured for securement of the cable in a manner which restrains movement of the cable; a cable connector protector coupled to the cable securing cartridge, the cable connector protector comprising at least one wall configured to shield a cable connector of the cable when the cable connector is plugged into a connector receiving port of the camera and when the cable protector device is attached to the camera; the cable securing cartridge comprising: walls defining a hollow passage space through which the cable can be passed for securement of the cable; a first constricted opening on a first side of the hollow passage space, the first constricted opening providing an entrance point for the cable to enter the hollow passage space; a second constricted opening on a second side of the hollow passage space, the second constricted opening providing an exit point for the cable to exit the hollow passage space; an internal cable constricting zone within the hollow passage space; and a cover configured to releasably attach over the hollow passage space, wherein the first constricted opening, the second constricted opening, and the cable constricting zone create a crooked passageway for the cable, when the cable is passed through the hollow passage space of the cartridge.
2. The cable protector device of claim 1, further comprising: an extension arm between the cable securing cartridge and the cable connector protector, wherein the extension arm positions a long axis of the cable connector protector in a substantial perpendicular alignment to a long axis of the cable securing cartridge.
3. The cable protector device of claim 2, the extension arm being configured to position the cable connector protector proximate the connector receiving port of the camera when the cable protector device is attached to an underside wall of the camera with the connector receiving port being on a side of the camera extending substantially perpendicular to the underside wall of the camera, such that the cable connector protector creates at least one shielding wall alongside the cable connector when the cable connector is plugged into the cable connector receiving port.
4. The cable protector device of claim 1, further comprising a mounting plate configured to mount to the camera, wherein the cable connector protector is an attachment component of the mounting plate, and wherein the cable securing cartridge is configured to releasably attach to the mounting plate.
5. The cable protector device of claim 1, wherein the cartridge comprises an elongated planar body.
6. The cable protector device of claim 1, wherein the cartridge includes: a bottom wall; a first side wall extending upwards from the bottom wall; a second side wall extending upwards from the bottom wall and opposite the first side wall; a front knob between the first and second side walls, the front knob forming a partial front wall with said first constricted opening provided between said first side wall and said front knob; a rear knob between the first and second side walls, the rear knob forming a partial rear wall with said second constricted opening provided between said first side wall and said rear knob; and an internal knob provided within passage space, said internal knob forming said internal cable constricting zone between said internal know and said second side wall.
7. The cable protector of claim 6 wherein cover is configured to releasably attach over the first and second side wall, and the rear, front, and internal knobs, opposite the bottom wall.

8. The cable protector device of claim 1, further comprising a mounting plate configured to mount to the camera, wherein the cable securing cartridge is configured to releasably attach to the mounting plate.
9. The cable protector device of claim 8, the cable securing cartridge further include a latched peg coupled to a lever arm, the latched peg configured to securely latch to the mounting plate, and to disengage from the mounting plate via a release force on the lever arm.
10. The cable protector device of claim 1, wherein the cable protector device prevents the cable connector from detachment and/or damage by preventing the cable from being pulled, while shielding the cable connector from impact.
11. The cable protector device of claim 1, wherein the cable is a USB cable and/or data cable.